

Ticket 1: Unable to Access Internet

User Report:

My internet stopped working this morning. I can't open any websites in Chrome or Edge. Other people in the office seem to be working fine.

Impact:

- Cannot browse websites
- Cannot access cloud applications
- Work interrupted

Resolution:

1. Testing ping to an external IP.

```
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 192.168.254.10: Destination host unreachable.
Reply from 192.168.254.10: Destination host unreachable.
Reply from 192.168.254.10: Destination host unreachable.
Reply from 192.168.254.10: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
PS C:\Users\T900\Documents> |
```

Analysis: The “Destination host unreachable” error suggests a Layer 3 routing failure, most likely caused by an invalid or unreachable gateway.

2. Checking IP configuration to see if DHCP correctly provisioned a valid Layer 3 configuration to the VM.

```
Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : lan
Description . . . . . : Red Hat VirtIO Ethernet Adapter
Physical Address. . . . . : 1A-98-C5-17-C6-C5
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . : Yes
IPv6 Address. . . . . : fd0e:7300:fac4:c8a9:7867:adf5:aa61:19e5(Preferred)
Temporary IPv6 Address. . . . : fd0e:7300:fac4:c8a9:d1be:6800:7145:25d(Preferred)
Link-local IPv6 Address . . . . : fe80::e3f5:d36d:68cd:50d8%8(Preferred)
IPv4 Address. . . . . : 192.168.254.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::685e:ddff:fe23:7d64%8
                             192.168.254.1
DHCPv6 IAID . . . . . : 102406341
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-DE-5F-46-1A-98-C5-17-C6-C5
DNS Servers . . . . . : fe80::685e:ddff:fe23:7d64%8
                        1.1.1.1
                        8.8.8.8
                        fe80::685e:ddff:fe23:7d64%8
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
```

Analysis: Output shows DHCP is disabled, meaning the system is using a static IP configuration with manually assigned network settings. This rules out DHCP issues and indicates the problem likely lies in the static configuration. The default gateway may be incorrect or not aligned with the active network, and the IP/subnet may not match the virtual network scope.

3. Restoring automatic network configuration by setting the adapter's IP configuration and DNS configuration to DHCP.

```

PS C:\WINDOWS\system32> netsh interface ip set address "Ethernet" dhcp
PS C:\WINDOWS\system32> netsh interface ip set dns "Ethernet" dhcp
PS C:\WINDOWS\system32> ipconfig /renew
Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : lan
    IPv6 Address. . . . . : fd0e:7300:fac4:c8a9:7867:adf5:aa61:19e5
    Temporary IPv6 Address. . . . . : fd0e:7300:fac4:c8a9:d1be:6800:7145:25d
    Link-local IPv6 Address . . . . . : fe80::e3f5:d36d:68cd:50d8%8
    IPv4 Address. . . . . : 192.168.64.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::685e:ddff:fe23:7d64%8
                                192.168.64.1

```

Analysis: DHCP is now enabled and the system has a valid lease. IPv4, DNS, and default gateway have been correctly assigned by the DHCP server (192.168.64.1), indicating restored network configuration and proper subnet alignment.

4. Final verification of network connectivity performed via ping and DNS resolution tests prior to closure.

```

[t900@WIN-148V3V0EN0M C:\Users\T900>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=33ms TTL=114
Reply from 8.8.8.8: bytes=32 time=24ms TTL=114
Reply from 8.8.8.8: bytes=32 time=26ms TTL=114
Reply from 8.8.8.8: bytes=32 time=22ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 33ms, Average = 26ms

[t900@WIN-148V3V0EN0M C:\Users\T900>nslookup 8.8.8.8
Server:      UnKnown
Address:     fe80::685e:ddff:fe23:7d64

Name:       dns.google
Address:    8.8.8.8

```

Analysis: Network connectivity successful. Lab is now concluded.

Conclusion: Static IP configuration was identified with DHCP disabled, resulting in incorrect or misaligned network settings and loss of connectivity. The issue was resolved by re-enabling DHCP, allowing the system to obtain a valid IP address, subnet mask, default gateway, and DNS configuration from the DHCP server (192.168.64.1). Connectivity was restored after verifying successful network communication through ping and DNS resolution tests.